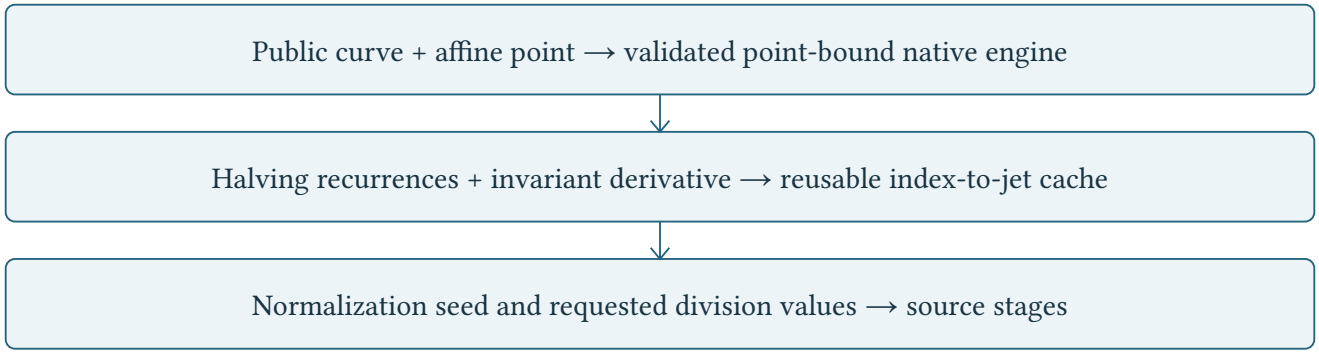
Native division values and normalization seeds

2026-11-03 · Public-point source component · Exact software checks

A dependency-free Rust core now computes division-polynomial values and their invariant-differential derivatives from a public curve and affine point. It preserves the normalization formula of the original sign source and retains one recurrence cache for subsequent requests at the same point.



Checked here: seeds / division values | Next: geometry + resultant

Construction. The derivation has D(x)=2y and D(y)=3x²+a. Value/derivative pairs obey the product rule, so the original even/odd division recurrences propagate both. Under the original torsion and coprimality premises, the seed solves c^(r²) D(psi_r)=-r. No scalar labels or precomputed signs enter this core.

Native evidence. At source21dce60cfc18, rustc1.99.1 built the standalone unit target and one invocation passed3 methods. The original generators over F353,F509,F701 reproduce seeds184,67,647 and all9 saved division values. Repeated requests leave the cache size unchanged. A separate Ward addition identity passes in both value and derivative on the fixed index ranges; explicit tangent, curve-membership and domain rejection controls also pass.

Reuse and cost. Each new division index is retained with its derivative; repeated requests use it. Original calls rebuilt independent caches. The native value-only interface still computes derivative data, and the map retains metadata and allocation costs. Repeated phase calls repeat the final inversion and exponentiation. No total-cost or elapsed-time saving is measured.

Scope. Prime p is a caller precondition; the backend supports p from5 through half the u64 maximum. Constructor checks nonsingularity, membership, canonical coordinates and nonzero y. This is a component port, not a completed elliptic sign source. No fresh recovery panel, cloud job or workspace Cargo suite ran.

Next. Port the point-grid and normalized recurrence/resultant stages; price value-only versus jet paths. Preserve the original full4,563-case cohort and admit its frozen native runtime before execution.

Evidence. lib.rs, fixtures.tsv, source/input hashes, projection script, check-receipt.txt and build/test logs. DERIVATION.md states contracts and limits. The byte-preserved original kernel is included for review, not execution.

Reference. Mathlib elliptic divisibility sequences, plus the preserved source kernel. Standard division recurrences and derivative rules are reused; no novelty or Lean certification is claimed. IACR fulltext fetch returned403 and is not used as evidence.